The Absolute Two-Ray Game as a Computational Testbed for the MMP

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Abstract

This note records a discussion about using the absolute two-ray game as a computational testbed for a general algorithm for the Minimal Model Program (MMP). The intended method does not use Cox rings or variation of GIT to solve the problem. Instead, it uses operations that plausibly extend to a general MMP algorithm: nefness testing, extremal-ray detection, construction of contractions by section rings, construction of flips by relative canonical algebras, and saturation in Rees-algebra or graph calculations. A terminal quartic threefold with a cA_5 point is proposed as a concrete example with two flips.

1 Absolute Picard rank two

Let X be a projective variety over a field of characteristic zero. In the absolute setting the base is $\operatorname{Spec} k$. If

$$\rho(X) = \dim_{\mathbb{R}} N_1(X)_{\mathbb{R}} = 2,$$

then $\overline{\mathrm{NE}}(X)$ is a closed convex cone in a two-dimensional real vector space. An ample divisor is strictly positive on nonzero effective curve classes, so $\overline{\mathrm{NE}}(X)$ is pointed. Consequently it has two boundary rays:

$$\overline{\mathrm{NE}}(X) = R_1 + R_2.$$

Thus Picard rank two really does reduce the numerical geometry to two rays.

There is nevertheless an important distinction between the following statements:

(i) the two boundary directions exist as real rays;

(ii) a boundary ray is rational;

(iii) it is generated by the class of an actual curve;

(iv) it is $(K_X + \Delta)$-negative and admits an extremal contraction.

For a general projective variety, a boundary ray can be irrational and need not contain a nonzero integral curve class. Under the hypotheses of the cone and contraction theorems, a $(K_X + \Delta)$-negative extremal ray is generated by a rational curve and admits a contraction.

The dual cone is

$$\mathrm{Nef}(X) = \overline{\mathrm{NE}}(X)^{\vee} \subset N^1(X)_{\mathbb{R}}.$$

Therefore computing the Mori cone and computing the nef cone are dual versions of the same problem. Picard rank two reduces the dimension, but it does not by itself reveal the slopes of the two boundary rays.

2 The Fano simplification

Assume that X is $\mathbb{Q}$-factorial and klt, that $\rho(X) = 2$, and that $-K_X$ is ample. Both boundary rays of $\overline{\mathrm{NE}}(X)$ are then K_X-negative. The cone theorem gives

$$\overline{\mathrm{NE}}(X) = \mathbb{R}_{\geq 0}[C_1] + \mathbb{R}_{\geq 0}[C_2],$$

where C_1 and C_2 may be chosen to be rational curves. One also has a length bound of the form

$$0 < -K_X \cdot C_i \leq 2 \dim X.$$

If mK_X is Cartier and $H = -mK_X$, then

$$1 \leq H \cdot C_i \leq 2m \dim X.$$

This turns the theoretical search for the two rays into a finite bounded-degree search in an anticanonical embedding. Such a Hilbert- or Chow-scheme search is still potentially expensive, but irrational boundary rays no longer occur.

Once $[C_1]$ and $[C_2]$ are known, nefness of a divisor D is reduced to two inequalities:

$$D \text{ is nef} \quad \Longleftrightarrow \quad D \cdot C_1 \geq 0 \ \text{ and } \ D \cdot C_2 \geq 0.$$

For a positive-dimensional Fano variety, K_X itself is of course not nef. The nontrivial nefness tests arise on extracted or flipped intermediate models, or for divisors other than K_X.

3 Recovering a ray from a nef threshold

Choose an ample divisor A and a direction $D \in N^1(X)_{\mathbb{R}}$. Define

$$\tau = \sup\{t \geq 0 \mid A + tD \text{ is nef}\}.$$

Then

$$L = A + \tau D$$

lies on the boundary of the nef cone. Since $\rho(X) = 2$, the orthogonal space

$$L^{\perp} \subset N_1(X)_{\mathbb{R}}$$

is one-dimensional and gives the corresponding boundary ray of $\overline{\mathrm{NE}}(X)$.

If a rationality theorem applies and an explicit denominator bound $q \leq Q$ is available for $\tau = p/q$, a nefness decision procedure can be used as a monotone oracle. Binary search to precision less than $1/(2Q^2)$, followed by rational reconstruction, determines τ exactly. The curve direction is then obtained by solving the single linear equation

$$L \cdot \gamma = 0.$$

This is one possible way in which Picard rank two can make an existing general nefness algorithm substantially more useful.

The numerical class γ is not yet an actual embedded curve. When L is semiample, one can construct the morphism defined by $|rL|$ for sufficiently divisible r and obtain curves from its positive-dimensional fibers.

4 A Cox-free computational MMP

The aim of the proposed experiment is not to compute the chamber decomposition of a Cox ring. Cox rings may be used externally to check an answer, but they should not be part of the algorithm under test. A generalizable step should instead have the following form.

Let Y_i be a $\mathbb{Q}$-factorial klt model of Picard rank two.

1. Use nefness testing and a one-dimensional threshold computation to identify a $(K_{Y_i}+\Delta_i)$-negative extremal ray R_i.
2. Find a supporting nef divisor L_i satisfying
$$L_i \cdot R_i = 0.$$
3. Construct the contraction from a section ring:
$$g_i \colon Y_i \longrightarrow Z_i = \operatorname{Proj} \bigoplus_{m\geq 0} H^0(Y_i, mL_i).$$
4. Compute the exceptional locus and decide whether g_i is of fiber type, divisorial, or small.
5. If g_i is small, construct the flip from the relative canonical algebra. If rK_{Y_i} is Cartier, use
$$Y_i^+ = \operatorname{Proj}_{Z_i} \bigoplus_{m\geq 0} g_{i*}\mathcal{O}_{Y_i}(mrK_{Y_i}).$$
6. Compute a presentation of Y_i^+, update divisor and intersection data, and repeat.

The difficult algorithmic questions are not merely the number of variables in the equations. They include certification that the relevant section or canonical algebra has been generated, exact detection of the extremal ray, and verification of the singularities after a flip.

5 Elementary examples and why they are insufficient

The blow-up of projective three-space at one point is a useful sanity check:
$$X = \operatorname{Bl}_p \mathbb{P}^3, \qquad p = [1:0:0:0].$$
In $\mathbb{P}^3_x \times \mathbb{P}^2_u$ it is defined by
$$x_1u_2 - x_2u_1 = x_1u_3 - x_3u_1 = x_2u_3 - x_3u_2 = 0.$$
Writing H for the pullback of a hyperplane and E for the exceptional divisor, one has
$$-K_X = 4H - 2E = 2H + 2(H-E),$$
and
$$\operatorname{Nef}(X) = \mathbb{R}_{\geq 0}[H] + \mathbb{R}_{\geq 0}[H-E].$$
The two contractions are the blow-down $X \to \mathbb{P}^3$ and the $\mathbb{P}^1$-bundle $X \to \mathbb{P}^2$. Thus either choice terminates in one step; no flip occurs.

This is typical of simple smooth Fano threefold examples. To obtain flips in dimension three, it is natural to allow terminal singularities and to start with a divisorial extraction from a Fano threefold of Picard rank one. The extracted model has Picard rank two and is the actual input to the absolute two-ray game.

6 A proposed quartic threefold test case

Consider the quartic threefold
$$X = X_{2,2} \subset \mathbb{P}^4_{x_0,\dots,x_4}$$
defined by

$$\begin{aligned} 0 &= (x_0x_1 - x_3^2)^2 - (x_0x_2 - x_4^2)^2 \\ &\quad + x_1^2x_3^2 + x_2^2x_4^2 + x_1^4 + x_2^4. \end{aligned}$$

It is a terminal factorial Fano threefold whose only singular point is

$$P = [1:0:0:0:0],$$

of type cA_5 [1].

Set

$$\begin{aligned} \alpha &= x_0(x_1 + x_2) - (x_3^2 + x_4^2), \\ \beta &= x_0(x_1 - x_2) - (x_3^2 - x_4^2). \end{aligned}$$

Up to harmless rescaling of α and β, the equation is

$$\alpha\beta + x_1^2x_3^2 + x_2^2x_4^2 + x_1^4 + x_2^4 = 0,$$

and the germ at P has the form

$$\alpha\beta + x_3^6 + x_4^6 + exthigher - weightterms = 0.$$

Take the discrepancy-one weighted blow-up with local weights

$$\mathrm{wt}(\alpha, \beta, x_3, x_4) = (3, 3, 1, 1).$$

Denote the extracted variety by
$$f\colon Y_0 \longrightarrow X.$$
Then $\rho(Y_0) = 2$. The known Sarkisov link has two flips and terminates in a degree-four del Pezzo fibration over $\mathbb{P}^1$ [1]. The local flips are hypersurface flips of type

$$(3, 1, 1, -1, -1; 2).$$

These facts should be treated as expected output for verification, not as input to the algorithm.

The extraction has a small weighted Rees presentation. With one exceptional parameter u, a convenient strict-transform presentation is

$$\begin{aligned} &u\alpha = x_0(x_1 + x_2) - (x_3^2 + x_4^2), \\ &u\beta = x_0(x_1 - x_2) - (x_3^2 - x_4^2), \\ &\alpha\beta + u^4(x_1^2x_3^2 + x_2^2x_4^2) + u^2(x_1^4 + x_2^4) = 0. \end{aligned}$$

Thus the initial quartic has five variables and one equation, while the extracted model can be represented using eight variables and three sparse equations. This is a plausible size for computation in systems such as Macaulay2 or Singular.

7 Expected computational behavior

The equations need not grow exponentially if every step is reduced to a minimal presentation. A local flip of type $(3,1,1,-1,-1;2)$ is a hypersurface calculation in five local variables. A global relative-Proj calculation may add one or two generators, but redundant old generators should then be eliminated immediately.

The following implementation choices are likely to cause an explosion:

- retaining the equations of every graph of every rational map;
- introducing new generators without eliminating old ones;
- replacing weighted or relative Proj presentations by large ordinary projective Veronese embeddings;
- computing section rings to a large degree in a single undifferentiated step;
- postponing all saturation and elimination until the end.

A better workflow is

$$\begin{aligned}\text{add minimal relative generators} &\longrightarrow \text{compute relations}\\ &\longrightarrow \text{saturate} \longrightarrow \text{eliminate redundant variables.}\end{aligned}$$

This is not a Cox-ring method; it is the ordinary algebraic implementation of Rees algebras, graph closures, and relative Proj.

The test can be staged as follows.

Stage 1. Supply the extremal ray and supporting divisor, and test only the construction of the two flips.

Stage 2. Supply the Picard and intersection data, but require the algorithm to discover the second ray using nefness tests.

Stage 3. Supply only the extracted model Y_0 and require automatic ray detection, contraction, smallness detection, flip construction, and singularity verification.

Stage 1 should be comfortably feasible. Stage 2 is also realistic. Stage 3 is a meaningful stress test for a general MMP algorithm; the likely bottleneck is certification of the relevant graded algebras and singularities rather than the raw number of equations.

8 Saturation by an exceptional or base ideal

Saturation is used to remove components introduced artificially by pullback, denominator clearing, or incidence equations.

Suppose a blow-up chart has exceptional divisor $E = \{u = 0\}$. If the total transform of an equation is

$$u^m \widetilde{f} = 0,$$

then its zero scheme contains the whole exceptional divisor as an extra component. The strict-transform ideal is

$$I_{\text{str}} = I_{\text{tot}} : (u)^\infty = \bigcup_{n \geq 0} (I_{\text{tot}} : (u)^n).$$

More generally,

$$I_{\text{str}} = I_{\text{tot}} : \mathcal{I}_E^\infty.$$

For example, blow up the origin of $\mathbb{A}^2$ and use the chart

$$x = u, \qquad y = uv.$$

The pullback of the curve $y - x^2 = 0$ is

$$u(v - u) = 0.$$

Saturation gives

$$(u(v - u)) : (u)^\infty = (v - u),$$

which is the strict transform. Its intersection with E is retained; only the unwanted component E itself is removed.

The same principle applies to graph constructions. If a rational function is represented by

$$s = \frac{f}{u^a},$$

then clearing denominators gives $u^a s - f = 0$, which may acquire spurious components on $u = 0$. Saturating by (u) extracts the closure of the graph over $u \neq 0$. More generally, for a rational map with base ideal J, one often computes

$$I_\Gamma = I_{\mathrm{inc}} : J^\infty.$$

The phrase "saturate by the exceptional ideal" is therefore shorthand. One should not blindly use the same ideal at every flip. The correct ideal is determined by the open locus whose closure is being computed: it may be an exceptional-divisor ideal, a base ideal, or an irrelevant ideal in a chosen relative-Proj presentation.

9 Conclusion

The absolute two-ray game is a promising intermediate test between toy examples and a general MMP implementation. Picard rank two removes branching in the numerical cone, while terminal quartic examples retain genuinely hard operations: extremal-ray detection, small contractions, flips, and singularity control. The cA_5 quartic above appears to have an appropriate balance: the input equations are sparse and small, but the expected computation contains two nontrivial flips. Keeping relative-Proj and Rees presentations minimal is essential for avoiding artificial equation growth.

References

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